

Python Study Group Gift

Case Study Project #13

Title: Market Price Forecasting System

Deadline: 24th July

A Welcome Note from Coach Timothy

Hello, Study Buddy! 🙌❤️

Welcome to your first **Case Study Project!**

First of all, I want to congratulate you for staying consistent. Every lesson you've completed and every project you've attempted is moving you one step closer to becoming a problem-solving Python developer.

One of the things I love most about this Study Group is that we don't learn Python in isolation. We learn Python through **real industry problems**.

Today's project takes us into an area where technology is making a huge difference:

Agriculture and Agricultural Market Intelligence.

Imagine you're a maize farmer.

You harvest your crops today.

Should you sell immediately?

Or should you wait another week?

How do you know if market prices are likely to increase or decrease?

Now imagine you're a food processing company purchasing thousands of bags of maize every month.

Buying too early could mean paying higher prices.

Buying too late could mean shortages or even higher costs.

These decisions affect millions of farmers, traders, businesses, and consumers every single day.

This is why agricultural organizations, commodity exchanges, cooperatives, and AgriTech companies build software to monitor and analyze market prices.

While advanced forecasting systems use Artificial Intelligence, historical data, and Machine Learning, they all begin with something much simpler:

- Collecting information
- Performing calculations
- Comparing values
- Evaluating conditions
- Presenting useful insights

And that's exactly what you'll do today.

Using only the Python concepts you've learned from **Lesson 1 to Lesson 5**, you'll build a simple **Market Price Forecasting System**.

Remember...

We're **not predicting the future**.

We're learning how software prepares information that helps decision-makers understand market trends.

Let's get started!

Industry Scenario

You have recently joined **AgriMarket Intelligence Ltd.** as a **Junior Python Developer**.

The company provides daily market intelligence reports to farmers, food processors, agricultural cooperatives, exporters, and government agencies.

Every morning, market officers collect information about commodity prices from different agricultural markets.

Your manager wants you to develop a simple Python program that analyzes today's market information and provides a summary for decision-makers.

Since you're still learning Python, your solution must use only the concepts covered in **Lessons 1–5**.

Learning Objectives

By completing this project, you should be able to:

- Create meaningful variables
 - Choose appropriate data types
 - Perform arithmetic calculations
 - Update values using assignment operators
 - Apply comparison operators
 - Combine business rules using logical operators
 - Present agricultural market information professionally
 - Think like a software developer building solutions for the agriculture industry
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Daily Market Information

The market officer provides the following information.

Information	Value
Commodity	Maize
Market Location	Kano Central Market
Yesterday's Price (per 100kg bag)	₦58,000
Today's Price (per 100kg bag)	₦61,500
Expected Market Price	₦60,000
Quantity Available (bags)	420
Additional Supply Received	80
Minimum Supply Requirement	450
Market Open	True

Create meaningful variables for all the information above.

Project Tasks

Task 1 — Create the Variables

Create variables for all the information provided.

Choose appropriate data types for each value.

Task 2 — Calculate the Price Difference

Calculate how much the commodity price changed from yesterday to today.

Formula:

$$\text{Price Difference} = \text{Today's Price} - \text{Yesterday's Price}$$

Store the result in a variable.

Display the answer.

Task 3 — Calculate the Percentage Price Change

Calculate the percentage increase in price.

Formula:

$$\text{Percentage Price Change} = (\text{Price Difference} \div \text{Yesterday's Price}) \times 100$$

Display the result.

Round the answer where appropriate.

Task 4 — Update Available Supply

Later in the day, another truck delivers **80 additional bags** of maize.

Update the available quantity using an **assignment operator**.

Display the updated quantity.

Task 5 — Check if the Expected Market Price Has Been Reached

Determine whether today's market price is **greater than or equal to** the expected market price.

Store the result in a Boolean variable.

Display the answer.

Task 6 — Check if Supply Is Sufficient

Determine whether the updated quantity available is **greater than or equal to** the minimum supply requirement.

Store the result in a Boolean variable.

Display the answer.

Task 7 — Determine if Market Conditions Are Favorable

The market is considered **Favorable** if:

- The market is open

AND

- The expected market price has been reached.

Use a logical operator to evaluate the condition.

Store the result in a Boolean variable.

Display the answer.

Task 8 — Determine if Market Monitoring Is Required

The market analyst wants additional monitoring if:

- Market conditions are **not favorable**

OR

- Supply is **not sufficient**.

Store the result in a Boolean variable.

Display the answer.

Task 9 — Generate the Market Price Forecast Report

Display your findings in a professional report similar to the example below.

```
=====
  AGRICULTURAL MARKET PRICE ANALYSIS REPORT
=====
```

Commodity: Maize

Market Location: Kano Central Market

Yesterday's Price: ₦58,000

Today's Price: ₦61,500

Price Difference: ₦3,500

Percentage Price Change: 6.03%

Updated Quantity Available: 500 bags

Minimum Supply Requirement: 450 bags

Expected Market Price: ₦60,000

Expected Price Reached: True

Supply Sufficient: True

Market Open: True

Market Conditions Favorable: True

Additional Market Monitoring Required: False

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Ensure your report is neat, readable, and suitable for farmers, commodity traders, and agricultural market analysts.

BONUS CHALLENGE (Optional)

Another market submits the following report.

Information	Value
Commodity	Rice
Market Location	Makurdi International Market
Yesterday's Price	₦ 72,500
Today's Price	₦ 70,000
Expected Market Price	₦ 71,000
Quantity Available	300 bags
Additional Supply	100 bags
Minimum Supply Requirement	450 bags
Market Open	True

Perform the same calculations and analysis for this market.

Then answer the following questions:

1. Which commodity recorded the larger price increase (or smaller decline)?

2. Which market has the higher current selling price?
 3. Which market reached its expected market price?
 4. Which market has sufficient supply after the update?
 5. Which market requires additional monitoring?
 6. If you were a commodity trader planning to purchase produce in bulk, which market would you choose based on your analysis? Explain your decision using the values calculated by your program.
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Project Rules

This project must be completed using **only** the concepts covered in **Lessons 1–5**.

You are expected to use:

- ✓ Variables
- ✓ Data Types
- ✓ Arithmetic Operators
- ✓ Assignment Operators
- ✓ Comparison Operators
- ✓ Logical Operators

Do **NOT** use:

- `if` statements
- Loops (`for` or `while`)
- Functions
- Lists
- Tuples
- Dictionaries
- Classes
- File handling
- Exception handling

The goal is to strengthen your understanding of Python fundamentals before we begin building programs that make automated decisions in future lessons.

Tips from Coach Timothy

Hey Study Buddy,

One of the biggest responsibilities of a developer is turning raw data into meaningful information.

As you complete this project, don't just think about writing code, think about the people who will use the results.

Ask yourself:

- Did I choose meaningful variable names?
- Are my calculations correct?
- Did I use assignment operators properly when updating values?
- Did I apply comparison operators accurately?
- Did I combine conditions correctly using logical operators?
- Is my report clear enough for a farmer, market analyst, or agricultural business manager to understand?

Remember:

Good developers write code. Great developers build solutions that help people make better decisions.



Submission Instructions

Submit the following:

- Your Python (.py) file or Jupyter Notebook (.ipynb)
- Double-check your calculations before submission
- Send a screenshot showing your program running successfully to the WhatsApp group, tag Coach Timothy
- Ensure your code is well-organized with comments where necessary
- Upload your work before the submission deadline shared in the Study Group
- Blog about this on your LinkedIn with snippet of your codes, tag Coach Timothy ([connect](#)) and recognize [SmartBizCrux](#)
- Use the hashtag #PythonStudyGroup #StudyGroup #CaseStudy
- Document your work on Github too (here is a template to guide you

Late submissions may be graded as 50% deduction.

Final Words from Coach Timothy

Study Buddy,

Agriculture is no longer driven by experience alone, it is increasingly driven by **data**. Farmers, cooperatives, commodity traders, food manufacturers, and government agencies all rely on timely market information to make smarter decisions.

Today's project introduced you to the foundations of agricultural market analysis. Even with only variables and operators, you've built a program that performs meaningful calculations, evaluates market conditions, and communicates valuable insights.

As we move deeper into the Python Study Group, you'll learn how to automate analyses like this for hundreds or even thousands of market records. But those advanced skills will only be possible because you've taken the time to master the fundamentals.

Never underestimate the power of consistency.

Remember our Study Group philosophy:

The best Python developers don't just learn syntax—they learn to solve real-world problems, one project at a time.

Enjoy!

— *Coach Timothy*